BES–H_2O_2–Ac

Hydrogen peroxide (H_2O_2) detection probes.

> Manufacturer : Wako
> Molecular Formula : $C_{28}H_{11}F_7O_8S$
> M.W. = 640.44 g / mol

Procedure

Step 1: Prepare stock solution (10 mM)

Dissolve 1 mg powder in **156.2 μL DMSO** to form a 10 mM stock solution. Store at room temperature, must be **protected from light**.

Step 2: Stain

Short time incubation

Dilute to **10 - 50 μM** with growth medium. Incubate at room temperature for ≈ 20 mins, protected from light.

Long time incubation

1. Make medium contains **100 nM** BES-H_2O_2-Ac.

- Add **4 μL** of 10 mM stock solution to **400 mL** medium before the medium solidified (~ 50°C), mix thoroughly, must be protected from light.

2. Grow the plant on this medium for ≈ 6 days, cut the root and observe using confocal microscope.

Step 3: Confocal parameters

Set the confocal parameters as follow:

Parameters	Theoretical value	Our machine
Excitation peak	485 nm	488 nm
Emission peak	515 nm	505 ~ 541 nm
Magnification		5X
Laser intensity (**488 nm**)		20.0%
Master Gain (**488 nm**)		700 V
Master Gain (**T-PMT**)		240 V
Digital Gain		1.0
Pinhole		≈ 35 μm
Z-stack interval		16 μm
Scan speed		6
Scan direction		↔

Step 4: Image processing

Open the CZI file with ImageJ, view stack with `Hyperstack`.

1. Generate BES and T-PMT layers

Repeat this process twice to generate both the BES and T-PMT layers:

`Image` ⟶ `Stacks` ⟶ `Z Project...` ⟶ Projection type: `Max intensity` ⟶ `File` ⟶ `Save As` ⟶ `Tiff...`

2. Generate stacks

Load in the previously generated BES and T-PMT TIFF files.

`Image` ⟶ `Stacks` ⟶ `Images to Stack`

3. Measure distance

Measure the distance between root tip and the BES signal distal boundary. The root tip can be easily recognized in the T-PMT layer.

Done !!!
